



Chardan Overview

March 2023



We Are A Full Service, Global Investment Bank

Where Disruption Banks

Founded in 2002 and headquartered in New York, Chardan is an independent, full-service, global investment bank focused on addressing the capital markets needs of public and private disruptive technology companies.

\$30 billion

**aggregate capital raised globally
since inception**



Transactional Expertise

Focused on novel M&A and capital markets solutions globally

Established leading SPAC advisory practice with over 20 years of transaction experience

Diverse sector experience and depth of knowledge

Full service – we provide advice from strategy to execution

First mover across sectors, strategies, and structures

Best-in-Class Research

Our independent, actionable insights power the hundreds of company and industry reports we publish each year

Over 80 companies under coverage

Majority of our analysts are ranked in the top third of ~8,300 Wall Street Analysts, and one of them is ranked in the top 2%

Extensive Network

Our robust relationships can be leveraged through hundreds of hosted and sponsored events

~3,000 investor interactions facilitated in 2022

Over 1,000 public and private companies participated in Chardan events in 2022

Host and sponsor of 7 conferences in 2022

Highly Specialized

Our institutional equities team is made up of professionals with long tenures and focused experience.

Approximately 400 dedicated accounts

Accounts include hedge funds, VCs, family offices, and mutual funds

Accounts span across the U.S. and EU



Full Suite of Chardan's Practice

Capital Markets

Partner with unique and disruptive companies to raise capital and help position them to institutional investors

Capital Raise

- IPOs, Follow-Ons (Primary and Secondary)
- PIPEs
- Private Placements
- ATMs

After Market Support

- Deal Roadshows
- Non-Deal Roadshows
- Corporate Access

Special Situations

- SPACs
- Restructuring of Existing Debt and/or Equity
- Bridge Financings
- Leveraged Transactions
- Reverse Mergers

Sales & Trading

Deliver sales, trading and market-making services for global institutions, hedge funds and industry innovators

Sales

- Research Recommendation Delivery
- Idea Events/Dinners
- Management Conference Calls

Trading

- Block Trades
- Direct-Access Solutions
- Equity/Derivative Trading
- Idea Generation
- Market Making

Prime Brokerage

- Wedbush Securities

Equity Research

Insightful equity research on domestic and international emerging growth companies

Sector Coverage

- Biotech & Pharmaceuticals
 - Autoimmune
 - Cell Therapy
 - CNS
 - Disruptive Oncology
 - Hematology
 - Gene Editing
 - Infectious Disease
 - Lipids
 - Microbiome
 - Ophthalmology
 - Rare Diseases
 - RNA Gene Therapy
 - Vaccines
 - Vector Gene Therapy
- Digital Health & Healthcare IT
- Medical Devices
- Diagnostics
- SPACs and Disruptive Technologies

Advisory

Offer objective advice to clients looking to fulfill their fiduciary duties and mitigate risk

Financial Advisory

- Sell Side Advisory
- Buy Side Advisory
- General Advisory
- Due Diligence Services

Transaction Opinions

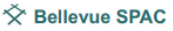
- Fairness Opinions
- Valuation Opinions
- Solvency and Other Capital Adequacy Opinions

Product Valuation

- Structured Product Valuation and Pricing
- Derivative Valuation



Selected Recent Transactions

 Intuitive Machines Inflection Point \$892 Million M&A February 2023 Capital Markets Advisor	 SOUTHLAND INDUSTRIES Legato II \$444 Million M&A February 2023 Capital Markets Advisor	 Bellevue SPAC Bellevue Life Sciences Acquisition \$69 Million IPO February 2023 Sole Book-Runner	 Allurion Allurion \$100 Million Chardan Equity Facility (ChEF) February 2023 Underwriter	 Orchestra BioMed Health Sciences 2 \$317 Million M&A January 2023 M&A and Capital Markets Advisor	 Cognos Therapeutics Nocturne Acquisition \$175 Million M&A January 2023 Capital Markets Advisor	 COYA THERAPEUTICS Coya Therapeutics \$16 Million IPO December 2022 Lead Book-Runner	 ALPHATIME ACQUISITION CORP AlphaTime \$69 Million IPO December 2022 Sole Book-Runner	 CARAVELLE Pacifico Acquisition \$544 Million M&A December 2022 M&A Advisor
 Wellous Kairous Acquisition \$323 Million M&A December 2022 M&A and Capital Markets Advisor	 TENAYA THERAPEUTICS Tenaya Therapeutics \$82 Million Follow On November 2022 Co-Lead Manager	 ATLAS Quantum FinTech \$75 Million M&A November 2022 M&A Advisor	 Grindr Tiga Acquisition \$1,822 Million M&A November 2022 Capital Markets Advisor	Alkaid Acquisition Alkaid Acquisition \$50 Million IPO Announced November 2022 Sole Book-Runner	 WAVETECH Welsbach Tech Metals \$263 Million M&A November 2022 M&A and Capital Markets Advisor	 NATURALSHRIMP Always Fresh • Always Natural Yotta Acquisition \$334 Million M&A October 2022 Financial Advisor	 RELIANCE Entertainment International Media \$140 Million M&A October 2022 M&A and Capital Markets Advisor	PERFECT Provident Acquisition \$1,022 Million M&A October 2022 Capital Markets Advisor
 DLT Blockchain Moon \$169 Million M&A October 2022 M&A Advisor	Hudson Acquisition I Hudson Acquisition I \$68 Million IPO October 2022 Sole Book-Runner	 dragonfly Chardan NexTech 2 \$470 Million M&A October 2022 M&A Advisor	Aquaron Acquisition Aquaron Acquisition \$54 Million IPO October 2022 Sole Book-Runner	 prestobio Ventoux CCM \$523 Million M&A September 2022 M&A Advisor	 LOGIQ Abri SPAC 1 \$146 Million M&A September 2022 M&A Advisor	 SENTI BIO Senti Bio \$50 Million Chardan Equity Facility (ChEF) September 2022 Underwriter	 Wag! CHW Acquisition \$381 Million M&A August 2022 M&A Advisor	 nauticus robotics CleanTech \$533 Million M&A September 2022 M&A Advisor
 TOMORROW Globalink \$310 Million M&A August 2022 Financial and Capital Markets Advisor	FiscalNote Duddell Street \$1,278 Million M&A July 2022 Capital Markets Advisor	 FR8tech Freight Technologies \$1.5 Million Convertible Preferred July 2022 Placement Agent	 ocugen Ocugen \$160 Million At-the-Market Offering June 2022 Placement Agent	 FORAFRIC MAROC Globis Acquisition \$273 Million M&A June 2022 M&A Advisor	 sky neos Neos / Sky UK M&A April 2022 Lead Financial Advisor	 NUVVE Nuvve \$25 Million At-the-Market Offering May 2022 Placement Agent	 YOTTA Yotta Acquisition \$115 Million IPO April 2022 Sole Book-Runner	 Redwoods Redwoods Acquisition \$115 Million IPO March 2022 Sole Book-Runner
A SPAC I Acquisition A SPAC I Acquisition \$69 Million IPO February 2022 Sole Book-Runner	 freapp HUDSON CAPITAL INC. FreightApp / Hudson Capital Inc. M&A February 2022 M&A Advisor	 ThinkMarkets ThinkMarkets \$10 Million Convertible Equity February 2022 Lead Financial Advisor	 BitFuFu Arisz Acquisition \$1,663 Million M&A January 2022 M&A Advisor	BroadCapital Broad Capital Acquisition \$101 Million IPO January 2022 Sole Book-Runner	 WTMAC Welsbach Tech Metals \$77 Million IPO December 2021 Sole Book-Runner	 Gardiner Healthcare Acquisitions Corp Gardiner Healthcare \$86 Million IPO December 2021 Sole Book-Runner	 GLOBALINK INVESTMENT Globalink Investment \$115 Million IPO December 2021 Sole Book-Runner	 PEAR THERAPEUTICS Thimble Point \$1,378 Million M&A December 2021 M&A Advisor



Chardan SPAC Overview



Chardan is the Highest-Performing Investment Bank Specializing in the SPAC Market

- **A leading SPAC Underwriter since 2016 with unparalleled client success**
 - Underwritten, led and co-managed 74 SPAC IPOs since 2016, raising over \$8 billion
 - Ranked top 3 in the league tables by deal count as sole-bookrunner
 - Involved in 122 SPAC IPO transactions since 2003
 - Chardan-advised teams achieve superior stock performance post-business combination
- **A best-in-class SPAC M&A Advisor since the early 2000's**
 - We have advised on 50 SPAC business combinations totaling over \$22 billion in value
 - Our strong relationships with key investors along with a proven legacy of creative, value-added solutions ensure that SPAC business combinations get closed
- Unparalleled sponsor experience - Chardan has sponsored/co-sponsored 15 SPACs, more than any other investment bank
- Decades of SPAC experience across a wide range of industry sectors
- **Chardan's dedicated SPAC advisory practice** is uniquely positioned to efficiently execute significant SPAC IPOs and business combinations on behalf of clients



Chardan's Long-Term Leadership in All Phases of SPAC Transactions



Top 25-Ranked SPAC Underwriters

Chardan's continues dominance in SPAC Underwriting, despite surge by bulge bracket competitors in 2020 & 2021

Rank	Underwriter	2016-2023 Sole Bookrunner	2016-2023 Total
1	Cantor	90	119
2	Chardan	47	74
3	Citigroup	44	174
4	Credit Suisse	43	127
5	Jefferies	41	74
5	EF Hutton	41	54
7	EarlyBird	40	83
8	Maxim	32	42
9	Goldman Sachs	25	115
10	Morgan Stanley	22	75
10	B. Riley	22	55
12	BTIG	20	49
13	I-Bankers	16	117
14	UBS	14	59
14	Deutsche Bank	14	87
16	BofA Securities	10	80
16	Cowen	10	34
18	Stifel	8	30
18	Wells Fargo	8	23
18	Ladenburg	8	49
21	J.P. Morgan	7	68
21	Oppenheimer	7	19
21	A.G.P.	7	8
24	Barclays	6	63
24	RBC	6	22

Top 25-Ranked SPAC M&A Advisors¹

Chardan is the top-ranked Buy-Side SPAC M&A Advisor by deal count in 2022

Rank	Advisor	2022	2018 - 2023 YTD Total	Sum of Deal Size (\$MM)
1	Chardan	16	36	\$17,289
2	Credit Suisse	7	25	\$52,038
2	Barclays	7	18	\$44,611
4	Morgan Stanley	6	22	\$84,180
5	Jefferies	5	23	\$28,781
6	Deutsche Bank	4	30	\$93,606
6	UBS	4	13	\$41,501
6	Cantor	4	9	\$11,963
9	Citigroup	3	20	\$46,200
9	Goldman Sachs	3	36	\$81,024
9	Evercore	3	4	\$9,922
9	William Blair	3	4	\$3,750
9	Cohen & Company	3	5	\$4,732
9	B. Riley	3	7	\$3,256
15	Cowen	2	8	\$14,518
15	Guggenheim	2	4	\$37,816
15	BTIG	2	10	\$19,673
15	Raymond James	2	5	\$3,195
19	J.P. Morgan	1	19	\$64,267
19	BofA Securities	1	16	\$43,847
19	Nomura	1	7	\$41,881
19	Moelis	1	4	\$7,391
19	RBC	1	4	\$9,146
19	Stifel	1	7	\$8,750
19	PJT Partners	1	2	\$3,478



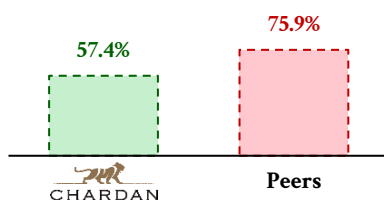
Chardan SPAC Performance Summary

IPOs¹

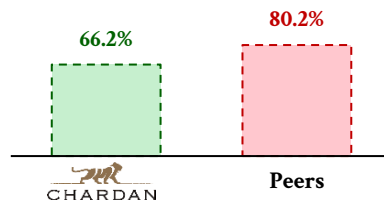
Higher prices and lower turnover reflect outsized demand for Chardan IPO's and signal investor confidence in longer-term value creation

Lead Underwriter	VWAP 1 Month Post-IPO
Chardan	\$10.19
Citigroup	\$10.17
EarlyBird	\$10.15
Network 1	\$10.15
Oppenheimer	\$10.10
BTIG	\$10.10
BofA Securities	\$10.06
J.P. Morgan	\$10.06
Morgan Stanley	\$10.05
Maxim	\$10.04
Ladenburg	\$10.04
ThinkEquity	\$10.03
SVB Securities	\$10.03
EF Hutton	\$10.01
UBS	\$10.01
Credit Suisse	\$10.01
Prime Number	\$9.99
Tiger Securities	\$9.99
Deutsche Bank	\$9.99
Needham	\$9.98
Cantor	\$9.94
I-Bankers	\$9.94

Average 1 Day Turnover

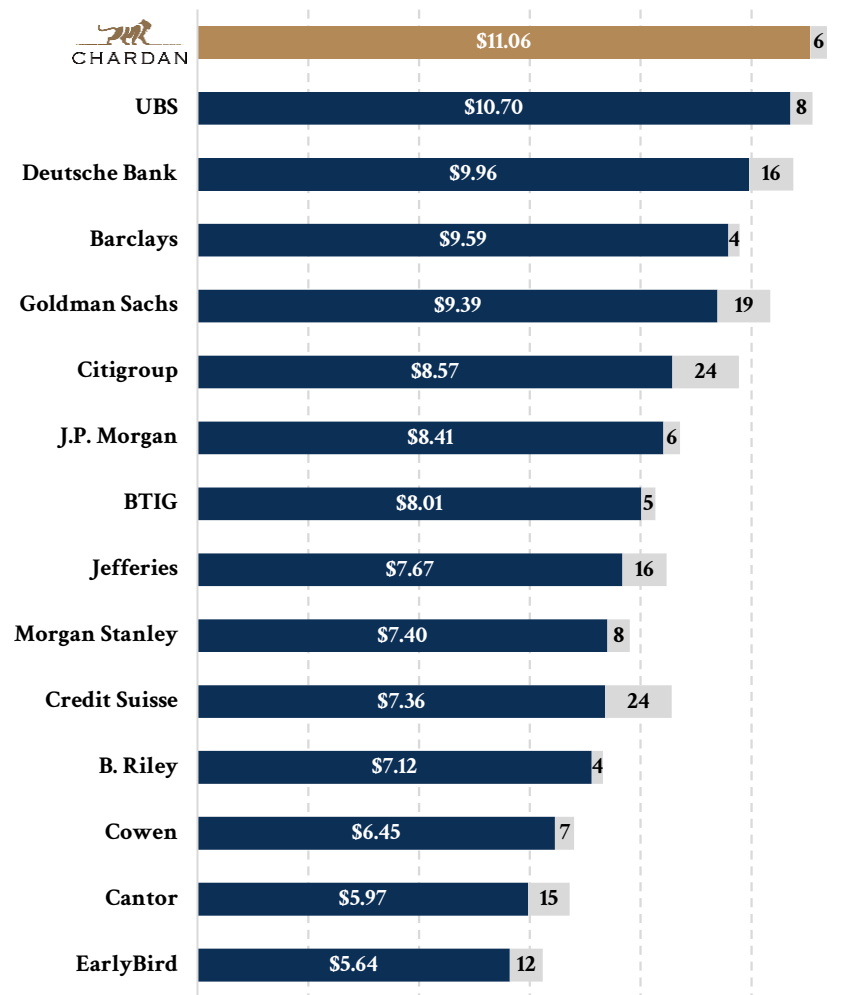


Average 1 Month Turnover



Business Combinations²

Chardan was the top-ranked SPAC underwriter by post-business combination performance in 2021



■ Average Stock Price ■ Number of Closed Deals

Notes: 1. Includes priced deals in the last 12 Months; as of 02/23/2023

2. Includes banks with at least 4 closed lead-left deals in 2021; Source: Chardan, FactSet prices as of 12/31/2021



Chardan's Underwritten SPACs have the Lowest Liquidation Rate

Percentage ¹ of SPAC liquidations by IPO vintage

Rank	Underwriter ²	2019	2020	2021	Average
1	Chardan	0.0%	18.2%	5.3%	7.8%
2	Stifel	N/A ³	0.0%	18.2%	9.1%
3	UBS	0.0%	15.4%	18.2%	11.2%
4	Deutsche Bank	0.0%	5.9%	33.3%	13.1%
5	EF Hutton	N/A ³	N/A ³	14.3%	14.3%
5	Oppenheimer	N/A ³	0.0%	28.6%	14.3%
7	Cantor	0.0%	23.8%	19.2%	14.3%
8	Goldman Sachs	0.0%	17.6%	29.3%	15.6%
9	Jefferies	0.0%	11.8%	36.4%	16.0%
10	Citigroup	0.0%	25.0%	27.5%	17.5%
11	Earlybird	16.7%	22.2%	13.6%	17.5%
12	Wells Fargo	N/A ³	25.0%	12.5%	18.8%
13	Credit Suisse	0.0%	31.4%	26.1%	19.2%
14	Cowen	0.0%	33.3%	25.0%	19.4%
15	Maxim	50.0%	25.0%	12.5%	29.2%

Notes: 1. Percentage is calculated as a number of liquidations divided by the number of total IPOs for that year by the underwriter

2. Includes underwriters which have completed at least 10 IPOs and have acted as sole book-runner or lead left on 50% of their IPOs

3. N/A indicates no IPOs underwritten in that year by that underwriter

Source: SPAC Analytics as of 02/22/23



Selected Chardan SPACs – Underwriter

Bellevue Life Sciences \$69 Million IPO February 2023 Sole Book-Runner	AlphaTime Acquisition \$69 Million IPO December 2022 Sole Book-Runner	Alkaid Acquisition \$50 Million IPO Announced November 2022 Sole Book-Runner	Hudson Acquisition I \$68 Million IPO October 2022 Sole Book-Runner	Aquaron Acquisition \$50 Million IPO October 2022 Sole Book-Runner	Yotta Acquisition \$115 Million IPO April 2022 Sole Book-Runner
Redwoods Acquisition \$115 Million IPO April 2022 Sole Book-Runner	A SPAC I Acquisition \$69 Million IPO February 2022 Sole Book-Runner	Broad Capital Acquisition \$101 Million IPO January 2022 Sole Book-Runner	Welsbach Tech Metals \$77 Million IPO December 2021 Sole Book-Runner	Gardiner Healthcare \$86 Million IPO December 2021 Sole Book-Runner	Globalink Investment \$115 Million IPO December 2021 Sole Book-Runner
Arisz Acquisition \$69 Million IPO November 2021 Sole Book-Runner	Mountain Crest Acquisition V \$69 Million IPO November 2021 Sole Book-Runner	Blockchain Moon \$115 Million IPO October 2021 Sole Book-Runner	Achari Ventures Acquisition \$100 Million IPO October 2021 Sole Book-Runner	Monterey Bio Acquisition \$115 Million IPO September 2021 Sole Book-Runner	WinVest Acquisition \$115 Million IPO September 2021 Sole Book-Runner
Pacifico Acquisition \$58 Million IPO September 2021 Sole Book-Runner	CHW Acquisition \$125 Million IPO August 2021 Sole Book-Runner	Chardan NexTech 2 \$127 Million IPO August 2021 Sole Book-Runner	Abri SPAC 1 \$58 Million IPO August 2021 Sole Book-Runner	International Media \$230 Million IPO July 2021 Sole Book-Runner	CleanTech Acquisition \$173 Million IPO July 2021 Sole Book-Runner



Selected Chardan DeSPACs – M&A / Capital Markets Advisor

<u>Inflection Point</u> Combines with  \$892 Million M&A February 2023 Advisor	<u>Legato II</u> Combines with  \$444 Million M&A February 2023 Advisor	<u>Health Sciences 2</u> Combines with  \$317 Million M&A January 2023 Advisor	<u>Nocturne Acquisition</u> *Combines with  \$175 Million M&A January 2023 Advisor	<u>Pacifico Acquisition</u> Combines with  \$544 Million M&A December 2022 Advisor	<u>Kairous Acquisition</u> *Combines with  \$323 Million M&A December 2022 Advisor
<u>Quantum FinTech Acquisition</u> *Combines with  \$75 Million M&A November 2022 Advisor	<u>Tiga Acquisition</u> Combines with  \$1,822 Million M&A November 2022 Advisor	<u>Welsbach Tech Metals</u> *Combines with  \$236 Million M&A November 2022 Advisor	<u>Provident Acquisition</u> Combines with  \$1,022 Million M&A October 2022 Advisor	<u>Yotta Acquisition</u> *Combines with  \$334 Million M&A October 2022 Advisor	<u>International Media</u> *Combines with  \$140 Million M&A October 2022 Advisor
<u>Blockchain Moon</u> *Combines with  \$169 Million M&A October 2022 Advisor	<u>Chardan NexTech2</u> Combines with  \$470 Million M&A October 2022 Advisor	<u>Ventoux CCM Acquisition</u> Combines with  \$523 Million M&A September 2022 Advisor	<u>Abri SPAC 1</u> *Combines with  \$146 Million M&A September 2022 Advisor	<u>CleanTech Acquisition</u> Combines with  \$533 Million M&A September 2022 Advisor	<u>CHW Acquisition</u> Combines with  \$381 Million M&A August 2022 Advisor
<u>Globalink Acquisition</u> *Combines with  \$310 Million M&A August 2022 Advisor	<u>Duddell Street</u> Combines with  \$1,278 Million M&A July 2022 Advisor	<u>Globis Acquisition</u> Combines with  \$273 Million M&A June 2022 Advisor	<u>Arisz Acquisition</u> *Combines with  \$1,663 Million M&A January 2022 Advisor	<u>Thimble Point Acquisition</u> Combines with  \$1,378 Million M&A December 2021 Advisor	<u>Trident Acquisitions</u> Combines with  \$504 Million M&A October 2021 Advisor



Selected Chardan SPACs – Principal

<u>Quantum FinTech Acquisition</u> *Combines with  \$201 Million IPO February 2021 \$75 Million M&A November 2022	<u>Chardan NexTech 2</u> Combines with  \$127 Million IPO August 2021 \$470 Million M&A October 2022	<u>Ventoux CCM Acquisition</u> Combines with  \$173 Million IPO December 2020 \$523 Million M&A September 2022	<u>CleanTech Acquisition</u> Combines with  \$173 Million IPO July 2021 \$533 Million M&A September 2022	<u>Globis Acquisition</u> Combines with  \$115 Million IPO July 2021 \$273 Million M&A June 2022
Gardiner Healthcare Acquisitions \$86 Million IPO December 2021 Sole Book-Runner	Monterey Bio Acquisition \$115 Million IPO September 2021 Sole Book-Runner	<u>Chardan Healthcare 2</u> Combines with  \$86 Million IPO April 2020 \$168 Million M&A September 2021	<u>LifeSci Acquisition</u> Combines with  \$66 Million IPO March 2020 \$137 Million M&A December 2020	<u>LifeSci Acquisition II</u> Combines with  \$80 Million IPO November 2020 \$1,300 Million M&A May 2020
<u>Chardan Healthcare</u> Combines with  \$70 Million IPO December 2018 \$248 Million M&A October 2019	<u>Chardan 2008 Acquisition</u> Combines with  \$55 Million IPO August 2008 \$212 Million M&A January 2010	<u>Chardan South China</u> Combines with  \$35 Million IPO August 2005 \$184 Million M&A January 2008	<u>Chardan North China</u> Combines with  \$35 Million IPO August 2005 \$192 Million M&A September 2007	<u>Chardan China</u> Combines with  \$24 Million IPO March 2005 \$151 Million M&A November 2005



Case Study: Inflection Point / Intuitive Machines (Nasdaq: LUNR)

Inflection Point Acquisition Corp.

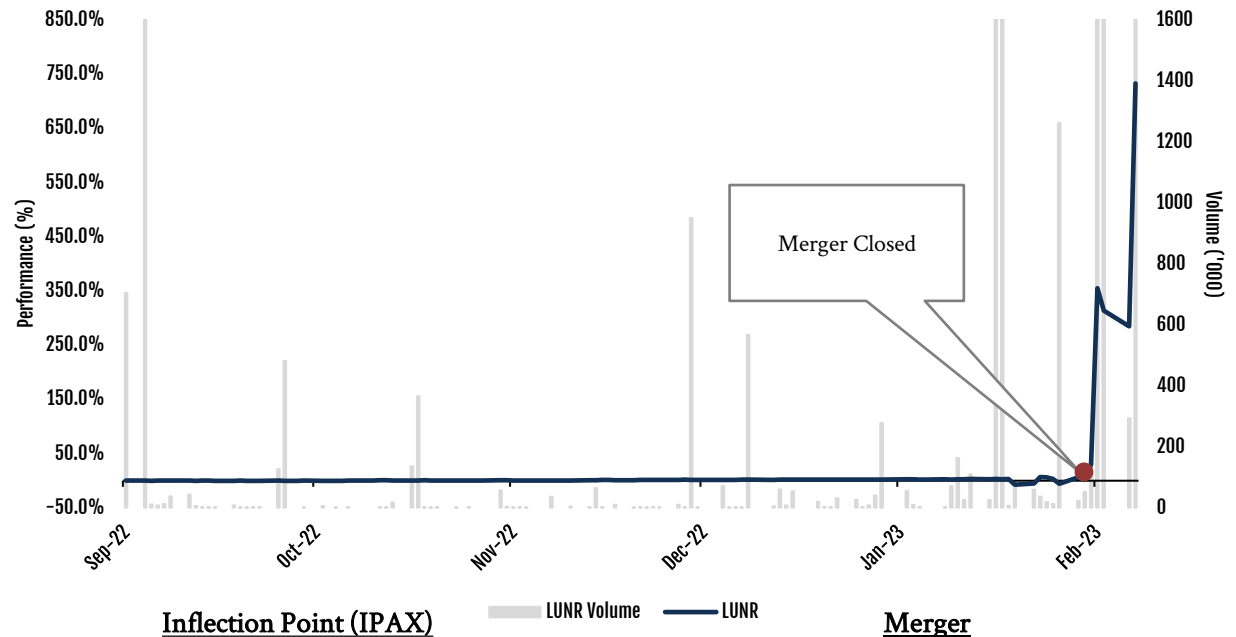
Combines with



\$892 million
February 2023
Chardan
Capital Markets Advisor

- Inflection Point closed its merger with Intuitive Machines in February 2023
- Intuitive Machines is an end-to-end space exploration company supplying space products and services to support sustained robotic and human transport to the Moon, Mars, and beyond.
- At the close of the merger, Inflection Point entered into the \$25M Forward Purchase Agreement ("FPA") agreement with certain investors to meet listing requirements. Post-merger, the stock price experienced a significant surge (reaching a high of \$136), allowing the company to unlock the FPA and access the \$25M in funding
- Chardan provided critical technical support so Inflection Point could meet stock exchange listing requirement and close their merger on time

Stock Performance from Announcement



Inflection Point (IPAX)	
IPO Date	09/22/21
Size	\$329M
Structure	1 common; 1/2 warrant
Amount in Trust	100%
Term	24 months
Risk Capital	\$6.9M

Merger	
Announcement Date	09/16/22
Closed Date	02/13/2023
Pro-Forma Equity Value	\$892M
Rollover Equity Value	\$700M
Pro-Forma Ownership	- LUNR: 79% - IPAX Holders: 6% - IPAX Sponsor: 9% - Other: 6%

Case Study: Health Sciences 2 / Orchestra BioMed (Nasdaq: OBIO)

Health Sciences 2 Acquisition Corp.

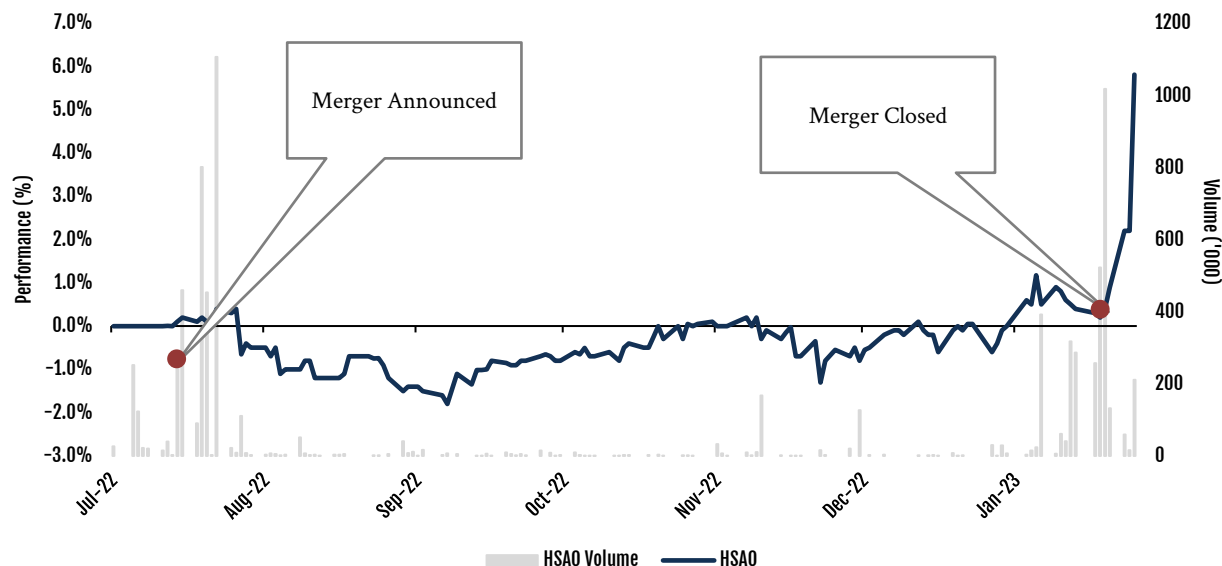
Combines with



\$317 million
January 2023
Chardan Lead Bookrunner
M&A Advisor
Capital Markets Advisor

- Orchestra BioMed is an innovative biomedical company partnered with leading medical device companies to accelerate high-impact device therapeutic technologies to patients
- \$70 million in gross proceeds from the business combination, including \$20 million in open market purchases by Medtronic and RTW, the sponsor of HSAQ, and additional backstop investments from RTW at closing
- The deal was announced alongside a new strategic partnership with Medtronic and \$110 million in Series D financing; including \$40 million from Medtronic, \$20 million from RTW, and investments from Perceptive Advisors, Terumo, and others
- Chardan was instrumental in advising HSAQ on deal structuring and capital markets strategy, including originating and executing the \$10 million purchase by Medtronic, resulting in low redemptions in a difficult market environment

Stock Performance from Announcement



Expanded Shareholder Base



Merger

Announcement Date	07/05/2022
Closed Date	01/26/2023
Pro-Forma Equity Value	\$317M
Trust Redemption	33.8%
Pro-Forma Ownership	- Orchestra: 44% - HSAQ Holders: 10% - HSAQ Sponsor and related parties: 30% - Medtronic: 16%



Case Study: Globis / Forafric (Nasdaq: AFRI)

Globis Acquisition Corp.

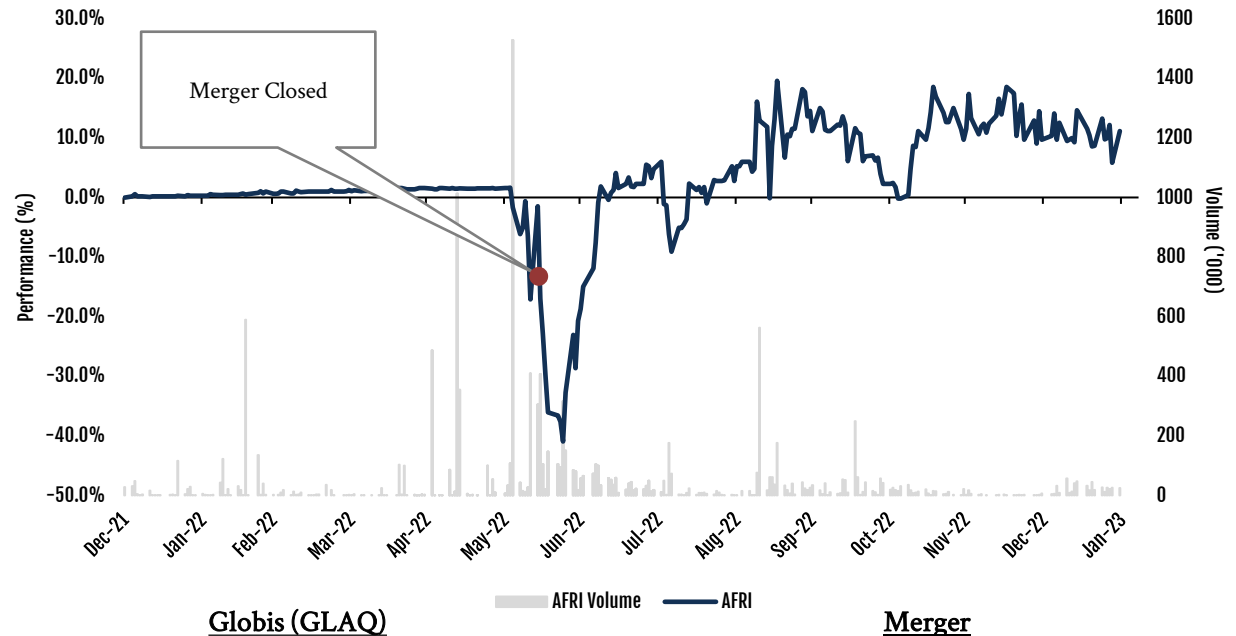
Combines with



\$273 million
June 2022
Chardan Sole Bookrunner
M&A Advisor

- Globis Acquisition Corp. closed its merger with Forafric Global PLC in June 2022
- Forafric is at the forefront of strengthening food systems, addressing Africa's food insecurity providing full agribusiness lifecycle operations with 10% of total market share in Morocco and distribution across 45 countries
- At the close of the merger, Forafric became the first ever Moroccan-based company to be listed on a U.S. exchange
- Chardan advised Globis on structural and regulatory elements of the merger, and further secured a \$15mn Forward Purchase Agreement for the company at closing, which was instrumental to meet NASDAQ listing requirements

Stock Performance from Announcement



IPO Date	12/11/2020
Size	\$115M
Structure	1 common; 1 warrant
Amount in Trust	101%
Term	12 months
Risk Capital	\$4.2M

Announcement Date	12/20/2021
Closed Date	06/10/2022
Pro-Forma Equity Value	\$273M
Rollover Equity Value	\$175M
Pro-Forma Ownership	- Forafric: 64% - GLAQ Holders: 7% - GLAQ Sponsor: 12% - PIPE and Other: 17%



Chardan Healthcare Overview



Why Chardan?

Premier Life Science Banking Platform

- **“Where disruption banks”**: Premier investment bank **addressing the needs of disruptive healthcare companies**, e.g. in genetic medicines (RNA, gene therapy, gene editing), regenerative medicines, precision medicines (oncology, cardiovascular, CNS, ophthalmology, autoimmune), microbiome and digital health
- Product pioneers with a **powerful combination of scientific and financial competence**
- Meaningful and **frequent connectivity to large- and mid-cap biopharma**, and disruptive biotech platforms
- Business model **built on establishing long-standing client relationships** from early stages to maturity
- **Established global platform** to optimize outputs of healthcare, M&A, scientific and clinical expertise

Robust Capital Markets Track Record

- **Raised upwards of \$1.7B in IPOs and \$3.5B in follow-ons** in Biotech as a bank since 2018; **also raised >\$600M in private capital** since 2015, spanning from Series A to crossover rounds
- Highly successful record with **key roles on numerous well-known genetic medicines IPOs** (Tenaya, Taysha, Passage Bio, Moderna, MeiraGTx, Solid Bio, AveXis, REGENXBIO) and **follow-ons** (Krystal Biotech, Moderna, CRISPR Tx, uniQure, Alnylam, AveXis)
- **Success driven by robust access** to large network of (1) fundamental **healthcare institutional investors**, (2) **science-focused investors in the US and Europe** and (3) **family offices and generalist funds**

Expansive Corporate Access Program

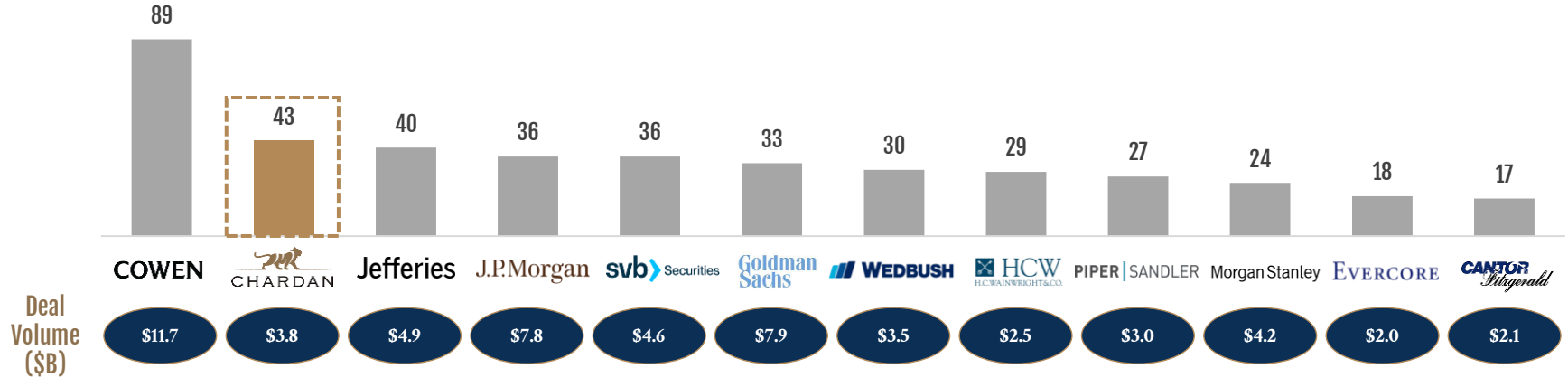
- Early, regular and **meaningful interactions post-IPOs**, supporting consideration for multiple follow-ons
- Proprietary **access to fundamental buy-side investors** via NDRs, KOL events and highly reputable conferences
 - Annual Genetic Medicines Conference, Genetic Medicines Manufacturing Summit and In Vivo Gene Therapy and Genome Editing Summit
- **Geographically diverse investor outreach** in the US and Europe, while catering to disruptive biotech expertise



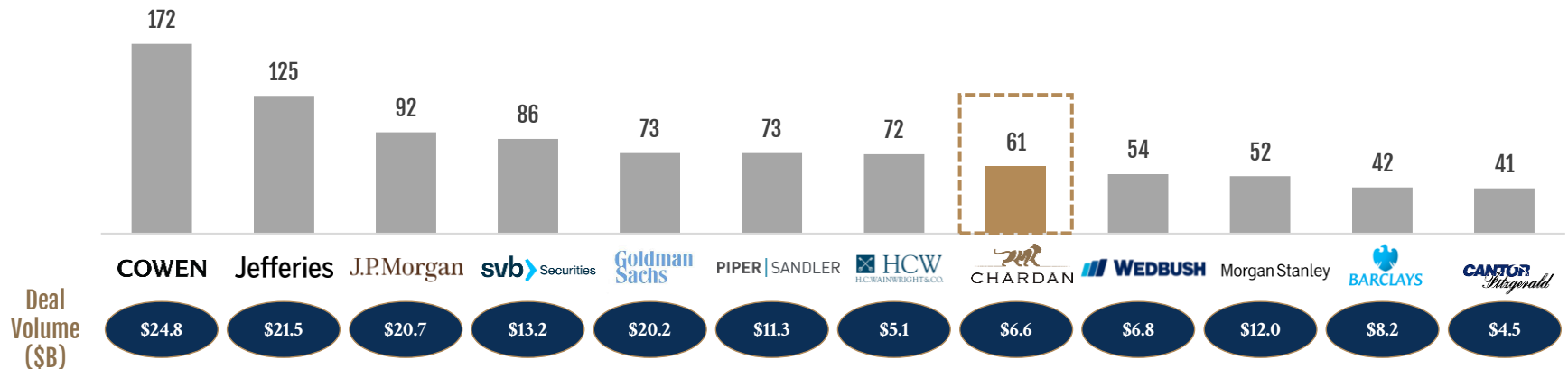
Chardan is a Leader in Genetic Medicines

Chardan has been one of the most active firms in public and private healthcare transactions in genetic medicines

In vivo Gene Therapy Ranking



All Genetic Medicines Ranking





Selected Recent Chardan Biotech Transactions

Notable \$145 Million Merger with VBL Feb 2023 M&A Advisor	SONNET BioTherapeutics \$15 Million Follow on Feb 2023 Lead Book-Runner	Orchestra \$317 Million SPAC Merger Jan 2023 Financial Advisor	COYA THERAPEUTICS \$17 Million IPO December 2022 Lead Book-Runner	TENAYA THERAPEUTICS \$75 Million Follow on November 2022 Co-Manager	SENTI BIO \$50 Million Chardan Equity Facility (ChEF) September 2022 Underwriter	PEAR THERAPEUTICS \$1,658 Million SPAC Merger December 2021 M&A Advisor	Krystal \$200 Million Follow On December 2021 Lead Manager
Better+ THERAPEUTICS \$187 Million SPAC Merger October 2021 M&A Advisor	SAB BIOThERAPEUTICS \$443 Million SPAC Merger October 2021 M&A Advisor	synlogic \$45 Million Follow On September 2021 Lead Manager	RENOVACOR \$177 Million SPAC Merger September 2021 M&A Advisor	TENAYA THERAPEUTICS \$207 Million IPO July 2021 Lead Manager	femasys \$34 Million IPO June 2021 Lead Book-Runner	FORGE BIOLOGICS \$120 Million Series B April 2021 Sole Placement Agent	neubase \$46 Million Follow On April 2021 Joint Book-Runner
ACHILLES THERAPEUTICS \$176 Million IPO March 2021 Lead Manager	4D pharma plc \$248 Million SPAC Merger March 2021 Financial Advisor	SOLID BIOSCIENCES \$144 Million Follow On March 2021 Lead Manager	PassageBio \$154 Million Follow On January 2021 Co-Manager	MEIRAGTx \$64 Million Follow On November 2020 Joint Book-Runner	immunome \$45 Million IPO October 2020 Joint Book-Runner	TAYSHA MULTIGENIC THERAPIES \$181 Million IPO September 2020 Lead Manager	IMMUNOVANT \$200 Million Follow On September 2020 Joint Book-Runner
Health Sciences Acquisitions 2 \$160 Million IPO August 2020 Lead Book-Runner	Krystal \$125 Million Follow On May 2020 Lead Manager	Chardan Healthcare 2 \$85 Million IPO April 2020 Sole Book-Runner	IMMUNOVANT \$139 Million Follow On April 2020 Joint Book-Runner	ReCode THERAPEUTICS \$80 Million Series A Merger with TranscripTx March 2020 Sole Placement Agent M&A Advisor	PassageBio \$248 Million IPO February 2020 Lead Manager	moderna \$500 Million Follow On February 2020 Co-Manager	CRISPR THERAPEUTICS \$315 Million Follow On November 2019 Co-Manager



Tenaya Therapeutics: Building Relationships With Industry Leaders



\$75 Million

Follow on
November 2022
Co-Manager



\$207 Million

IPO
July 2021
Lead Manager

Situation

- Tenaya Therapeutics (NYSE: TNYA) is a preclinical-stage biotechnology company based in South San Francisco, using a multi-modality approach including gene therapy, cellular regeneration, and precision medicine to develop treatments for cardiovascular diseases
- Chardan built a relationship with Tenaya stemming from the days of REGENXBIO, where Tenaya's CEO served as a Chief Business Officer
- In 2019, Chardan took the view that cardiac is an emerging tissue for genetic medicines
- Chardan's scientific expertise within its healthcare banking platform uniquely enabled it to advise Tenaya on its positioning relative to peers in its technology landscape
- Chardan provided advisory services ranging from scientific advisory to preparation of materials for board meetings
- Tenaya presented in Chardan's Genetic Medicines Conference 2020, as well as Chardan's Genetic Medicines Manufacturing Summit 2021
- Tenaya priced an upsized \$207M IPO in Jul-2021 with Chardan acting as lead manager
- In addition, Chardan helped with \$75M follow on raise in Nov-2022 acting as co-manager

Deal Background

- Chardan's transactional expertise from early-stage private financings through public market offerings matched the Company's needs to accelerate to an IPO less than 5 months after its last private financing round
- The opportunity arose for Chardan to execute this transaction due to its extensive work in the gene therapy sector including in names focused on cardiovascular disease
- In addition to transactional execution, Chardan's role as lead manager included heavy participation in scientific due diligence, S-1 drafting and preparation of presentation materials



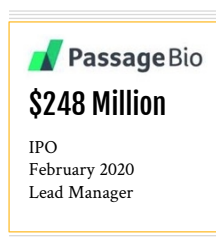
Passage Bio: Game Changing Early Stage Financings

Situation

- Passage Bio (NasdaqGS: PASG), a Philadelphia-based biotechnology company pioneering a novel *in vivo* gene therapy platform; formed in 2017 with Chardan's involvement
- Company later approached Chardan, along with a lead investor, to assemble an investor syndicate for its Series A financing; at the time, Passage had multiple preclinical gene therapy programs targeting rare monogenic diseases of the central nervous system
- As the sole placement agent on the Series A, Chardan:
 - Created a tailored recommendation for investor outreach based on the objectives of the Company and its lead investor
 - Efficiently and seamlessly organized a roadshow to introduce the Company to appropriate new investors
 - Managed the follow up diligence and data room processes
 - Maintained a consistent dialog with target investors to assess and consolidate sentiment; capital markets insight was provided to Company management regularly
 - Chardan Healthcare Investments co-invested alongside the syndicate
 - The Company announced a successful upsized, oversubscribed financing in Feb-2019
- Chardan also acted as a financial advisor in the company's second successful private financing
- Passage Bio filed for a \$248M IPO in Feb-2020; Chardan acted as lead manager
- Chardan also participated as co-manager for the Company's \$154M Jan-2021 follow-on offering

Deal Background

- The opportunity arose for Chardan to execute this transaction due to its extensive work in the gene therapy sector and strong relationships with early stage genetic medicine investors
- Top investors who participated in the Series A round included OrbiMed, Frazier, New Leaf Ventures and Versant
- The Series B included new investors Boxer Capital of Tavistock Group, Highline Capital Management, Logos Capital and Sphera Funds Management





MeiraGTx: Follow-On Offering Leadership Post-IPO

Leading Towards a Successful IPO

- MeiraGTx (NasdaqGS: MGTx) is a clinical-stage gene therapy company that, at the time, had four ongoing clinical programs and a broad pipeline of preclinical and research programs
- Chardan served as a placement agent in the \$85M Series C (crossover). With an initial focus on the eye, salivary gland and central nervous system, MeiraGTx used the funds to develop additional gene therapy treatments for patients suffering from a range of serious diseases
- Chardan Healthcare Investments also co-invested alongside Perceptive, Adena Estate, Alexandria Life Science Ventures, Seventh Avenue Investments, Aigle Healthcare Partners, EssexBio, Maximus Investments, Albion Capital, Cormorant, Amici Capital and Pontifax
- After a successful crossover round, the Company also approached Chardan to act as a lead manager in its eventual IPO
 - Chardan worked closely with the Company to arrange European TTW meetings and a manufacturing tour in London ahead of the IPO
 - European efforts resulted in a top five order

Post-IPO Support and Success

- Following the IPO, Chardan has continued to support the Company and acted as a co-placement agent on the \$80M private placement and joint book-runner on the \$75M follow on
- Most notably, Chardan has conducted meaningful marketing work with MeiraGTx in Boston/New York NDRs ahead of each deal
- Chardan provided conference support with showcase presentations and high quality 1x1 meetings at the Chardan Genetic Medicines Conference in 2018, 2019 and 2020, in addition to supporting marketing activities during the 2019 and 2020 JPM Healthcare Conference weeks
- Chardan facilitated participation of a trillion-dollar European Union-based fund in the company's follow-on offering



\$74 Million

Follow On
November 2020
Joint Book-runner



\$75 Million

Follow On
August 2019
Joint Book-runner



\$80 Million

PIPE
February 2019
Co-Lead
Placement Agent



\$75 Million

IPO
June 2018
Lead Manager



\$85 Million

Series C
April 2018
Placement Agent



Chardan Corporate Access



Well-structured Program to Connect with Potential Investors

Chardan has a large investor network

- Fundamental healthcare institutional investors
- Deep science investors
- Unique distribution platform of family offices and generalist funds

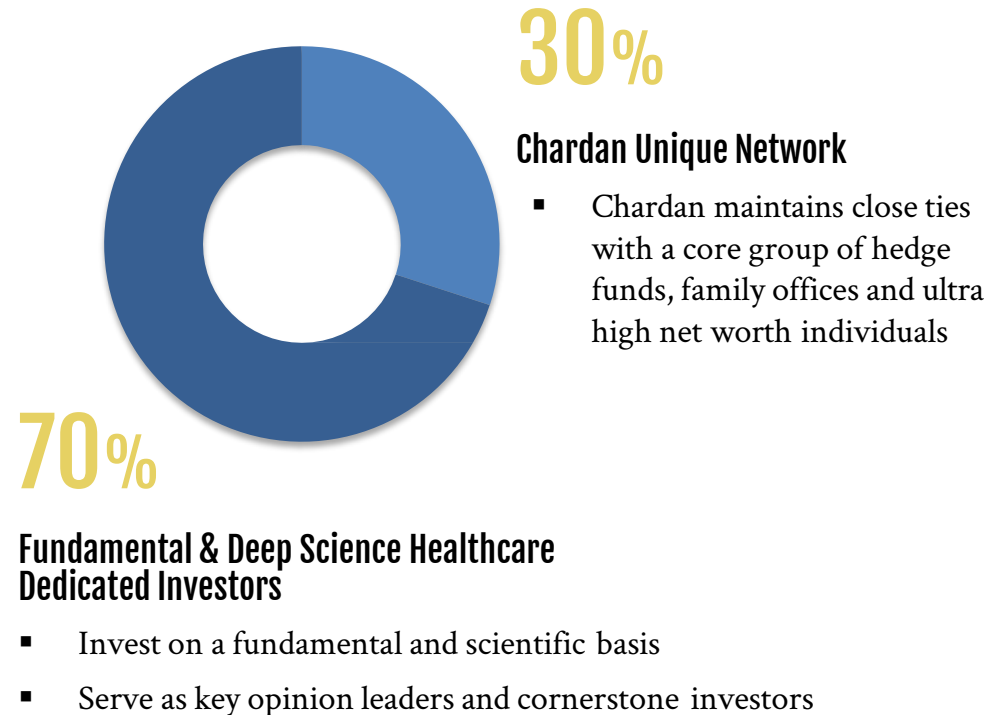
Event Types

- Non-deal roadshow
- KOL events
- Leadership calls/webinars
- Conferences
- Fieldtrips

Investor Outreach Objectives

- Introduction to new potential investors
- Identify potential high probability investors for open market purchases and potential offerings
- Identify any investor concerns and refine investment stories accordingly

Investor Outreach Breakdown





Investor Outreach

- Chardan is adept at laying the groundwork for successful investor interactions and can help you understand the effectiveness of a completed event

Pre-Event

- Shareholder analysis
- Analysis of research readership of company and peers
- Pre-qualification of investors through dialogue
- Host leadership Q&A call to prime the market

Post-Event

- Chardan follows up with each participating investor
- Poll investors on a specific element of your business about which you are trying to gauge market views
- Post-mortem call with management to discuss potential adjustments to messaging and changes in investor sentiment

Sample Feedback

President & CEO	Chief Science Officer	Investor Relations
Name	Name	Name

Investor A

"Very open and transparent on call. I would like to also meet the CEO in the future. Seems to be a better approach than Competitors' [anonymizing] one size fits all strategy, which seems dangerous. Was not aware of the best price rule for the government, seems that there is a long way to go for viable pricing."

Investor B

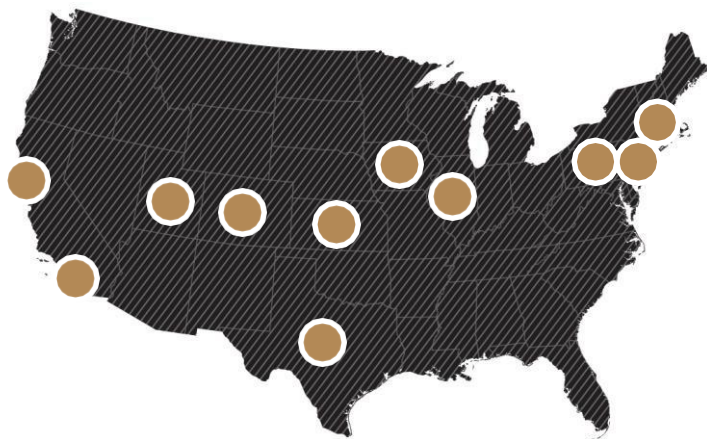
"I look forward to seeing further data at ASH, would give confidence. Insurers do not care about holistic costs, etc. I am not sure they can achieve the target they are announcing. Must tackle the issue of identifying patients, which will not be an easy challenge. Prefer that they are not focused on stock performance, taking a leadership position instead of thinking about the market."

Investor C

"Looking for disruptive innovative technologies, this seems to be a very good fit for the fund, Recruiting patients, durability and pricing are all still unknown, however company is well positioned. Would the company raise capital near term? Would be interested in follow up discussions."



Chardan's U.S. Healthcare Equities Distribution Platform



San Francisco

- 5AM Ventures
- Acuta
- Balyasny
- BVF
- Columbia Management
- EcoR1 Capital
- Farallon Capital
- Forward Management
- Franklin
- Kearny
- Lilly Asia Ventures
- NEA
- Novo A/S
- Palo Alto
- Partner Fund
- Redmile
- RS Investments
- Sofinnova Ventures
- Susquehanna
- Versant Ventures
- Vivo Capital

Los Angeles/ San Diego

- Capital Research
- Capital World
- DAFNA
- Invesco
- Nicholas Investment Partners
- PRIMECAP
- Tavistock
- Wall Street Associates

Denver / Kansas City / Salt Lake City

- American Century
- Arrowpoint
- Invesco
- Janus
- Waddell & Reed
- Wasatch

Dallas

- AWH
- Bain Capital
- Carlson
- Cloverdale
- Falcon Fund
- Gray's Creek
- HBK
- Highland
- Highside
- Hodges
- Ikarian
- Pinnacle
- Prosight
- Surveyor
- SWK Holdings
- White Rock

Midwest / Chicago

- Alyeska
- Artisan Partners
- Columbia Wanger
- Driehaus
- First Light Asset Management
- Principal Financial
- UBS Global AM

Mid-Atlantic

- Emerald Advisors
- Rock Springs
- T. Rowe Price
- Turner Investments

Boston

- Adage
- BlackRock
- Boston Company
- Brookside
- Camber Capital
- Columbia Mgt
- Cormorant
- Essex
- Eventide
- F-Prime
- Fidelity
- Granite Point Capital
- Longitude
- MFS
- Opaleye
- Putnam
- RA Capital
- State Street
- Tekla
- Wellington

New York

- 683 Capital
- Aisling
- Affinity
- Akkad Family Office
- AllianceBernstein
- Arrowgrass
- Avoro Capital
- Baker Brothers
- Broadfin
- Casdin Capital
- Citadel / Surveyor
- D.E. Shaw
- Deerfield
- Falcon Edge
- Federated
- Foresite
- Fosun
- Fred Alger
- Ghost Tree Capital
- Gilder Gagnon
- Great Point Partners
- GSAM
- Healthcor
- Invus
- J Goldman
- Jennison
- JP Morgan AM
- Kepos
- Lord Abbett
- Maven
- Marshall Wace
- Matrix Capital
- Millennium
- New Leaf
- Nexthera
- OrbiMed
- Perceptive
- Point72
- PointState
- Pura Vida
- QVT
- Ridgeback
- RTW
- Sabby Capital
- Sarissa
- Scopia
- Sio Capital
- TIAA-CREF
- Tri Locum
- UBS O'Connor
- venBio
- Venrock



Extensive European Distribution Capabilities

Targeted European Investor Hubs

Key Relationships in:

London

- 4Bio
- Abingworth
- AXA
- Arix
- Belsize
- GLG
- Eaton Vance
- Endurant
- IMPAX
- Marshall Wace
- Norges
- Point72
- Polar
- SV



Edinburgh

- Baillie Gifford

Stockholm

- Arctic Investment

Zurich

- BB Biotech
- Bellevue
- GAM
- HBM
- Keiger
- Pictet
- UBS
- Umblin

Paris

- Amundi
- Arbevel
- BNP
- Carmignac Gestion
- Sofinnova France
- Truffle Capital

Amsterdam

- AES Capital
- EQT Partners

Frankfurt

- ACEVO
- APUS
- Commerzbank
- Deka
- Deutsche Bank
- Union

Roadshow Capabilities

- Chardan actively helps biotech clients market to European investors with over 30 NDRs conducted per year
- Our roadshows offer our buy-side clients the opportunity to hear about new opportunities from senior management
- Active research dialogue in Europe complements the marketing activities of many of our clients
- Chardan has conducted hundreds of investor interactions in 2021 through virtual Non-deal Roadshows (vNDRs)

Recent Successful European Roadshows & vNDRs

Cabaletta Bio™

Virtual, Global

GenSight
BIOLOGICS

Virtual, Global

PassageBio

Virtual, Global

Sensorion
The Inner Ear Diseases company

Virtual, Global

4D
pharma plc

Virtual, Global

TAYSHA
GENE THERAPIES

Virtual, Global

BiomX

London

ProQR

Zurich

moderna

London



Equity Research Platform

SPACs and Disruptive Technologies Coverage

TMT / FinTech

Wag!

nauticus
robotics

FR8tech

FREIGHTOS

presto

PLAYBOY

Bmt

Electric Economy

wallbox

OUSTER

NUVE

Valens

Energy Storage



ENERGY VAULT
Enabling a Renewable World

dragonfly[®]
ENERGY
Limitless green energy

Solid Power

ESS
TECHNOLOGY

Li-Cycle[®]

Crypto



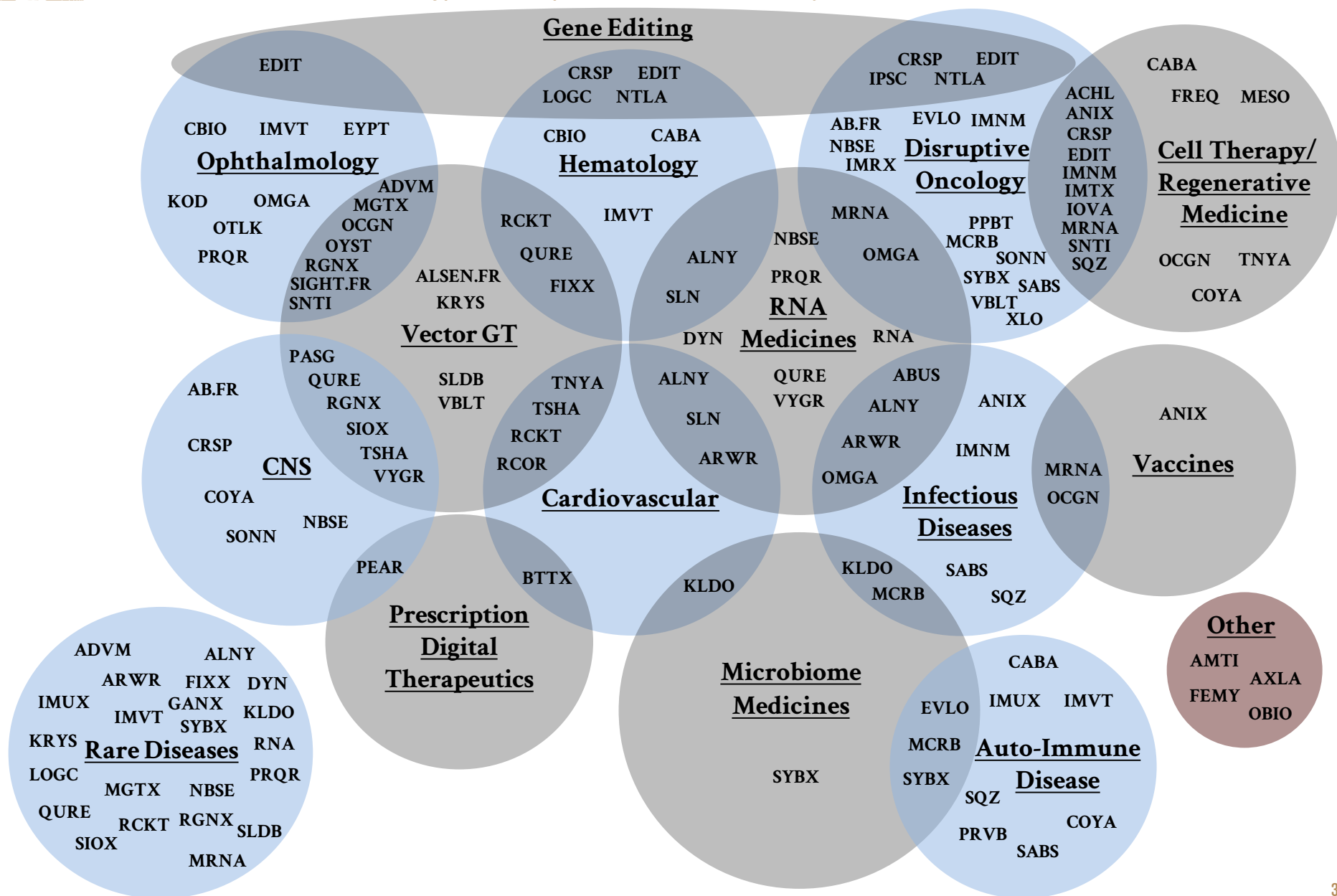
MARATHON
DIGITAL HOLDINGS

CLEANSARK



Premier Healthcare Coverage with Overlapping Verticals

Chardan looks to cover names in areas with strong potential to outperform healthcare in the next 5-10 years





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